

Simulated Users vs. Humans in Conversational Schema Construction

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May 21, 2026

Headline Claim

LLM-simulated users diverge from real human users during conversational schema construction in measurable, predictable ways. This project maps the divergence axes — where simulated users match humans and where they break.

Research Questions

RQ1. On which edit types do sim-users diverge most from humans?

Hypothesis: divergence concentrates on merge and delete — sim-users over-merge and under-delete.

RQ2. Do sim-users converge faster and to more-merged schemas than humans?

Hypothesis: sim-users finish in fewer turns with more merging.

Study Design — Match-Only

- Every session starts from **one fixed, deliberately messy seed schema** — duplicate nodes, wrong merges, over-splits.
- Only manipulated factor: **agent type** (human vs. sim-user), between-subject.
- Same task for all: edit the schema via an assistant-proposal turn loop (add / merge / split / rename / delete) until marked Done.
- Single session per participant; hard turn cap as backstop.

Corpus. TheMealDB recipe dataset — name, cuisine, category, tags, ingredients, instructions.

Sample size.

- Humans: target $N = 20$.
- Sim-users: 120 sessions on the fixed seed schema.

State of the Art — What is Relevant

- **Park et al. (2023, 2024).** Generative agents simulating human behavior — fidelity precedent.
- **Argyle et al. (2023); Aher et al. (2023).** LLMs as simulated human samples and experimental subjects.
- **“Sim-Users are Unreliable Proxies” (2026).** Closest precedent — sim-users systematically diverge.
- **Pan et al. (2025) — SCHEMA-MINER.** Human-in-the-loop schema mining; the expert loop we swap with a sim-user.
- **Zhang & Soh (2024) — EDC.** Extraction backbone. Not under study.

Open Questions

- Exact seed-schema content — which recipes, how much planted mess.
- Turn / step budget per session — the hard-cap value.
- Divergence metrics and per-RQ statistical tests — pending further research before the main experiment.